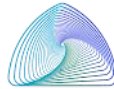


KINGDOM OF SAUDI ARABIA  
Ministry of Higher Education  
Taibah University  
College of Computer Science and  
Engineering (Girls Section)

جامعة طيبة  
TAIBAH UNIVERSITY



المملكة العربية السعودية  
وزارة التعليم العالي  
جامعة طيبة  
كلية علوم وهندسة الحاسب الآلي

# Airport Management System

A Project Report Submitted to Fulfill the Requirements of CS372

Group (4)

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03 November 2025

# Abstract

The Airport Management System helps manage airport activities more easily and efficiently. It stores and organizes information about flights, passengers, airlines, and staff. The system handles flight schedules, ticket bookings, check-in, and baggage tracking. It also helps monitor arrivals, departures, and delays to keep operation run smoothly. By using one system for all airport processes, it saves time, reduces errors, and improves the travel experience for passengers.

**Keywords:** Airport; Flights; Airlines; Management System; Baggage

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## **Chapter 1: Introduction**

### **1.1 Overview**

- Airports handle a large amount of critical data daily, including information related to flights, passengers, baggage, gates, and staff. Inefficient handling of this data can lead to booking conflicts, delayed flights, misplaced luggage, and overall operational inefficiency.
- The goal of this project is to design a database system that efficiently manages and organizes airport data. The proposed system will help maintain accurate and real-time information about flights, passengers, staff, security checks, gates, and baggage.
- The system provides effective tools for flight scheduling, seat allocation, gate identification, baggage tracking, and staff management. Through a database and well-defined relationship among entities. Also enhances data consistency, supports decision-making, minimizing duplication errors and improves the overall efficiency of airport operations.

### **1.2 Client Context**

- The Clients are the airport management staff and passengers. The airport staff need system to manage flights, baggage and more.
- Passengers need system to help them book tickets, check in and many more airport services.

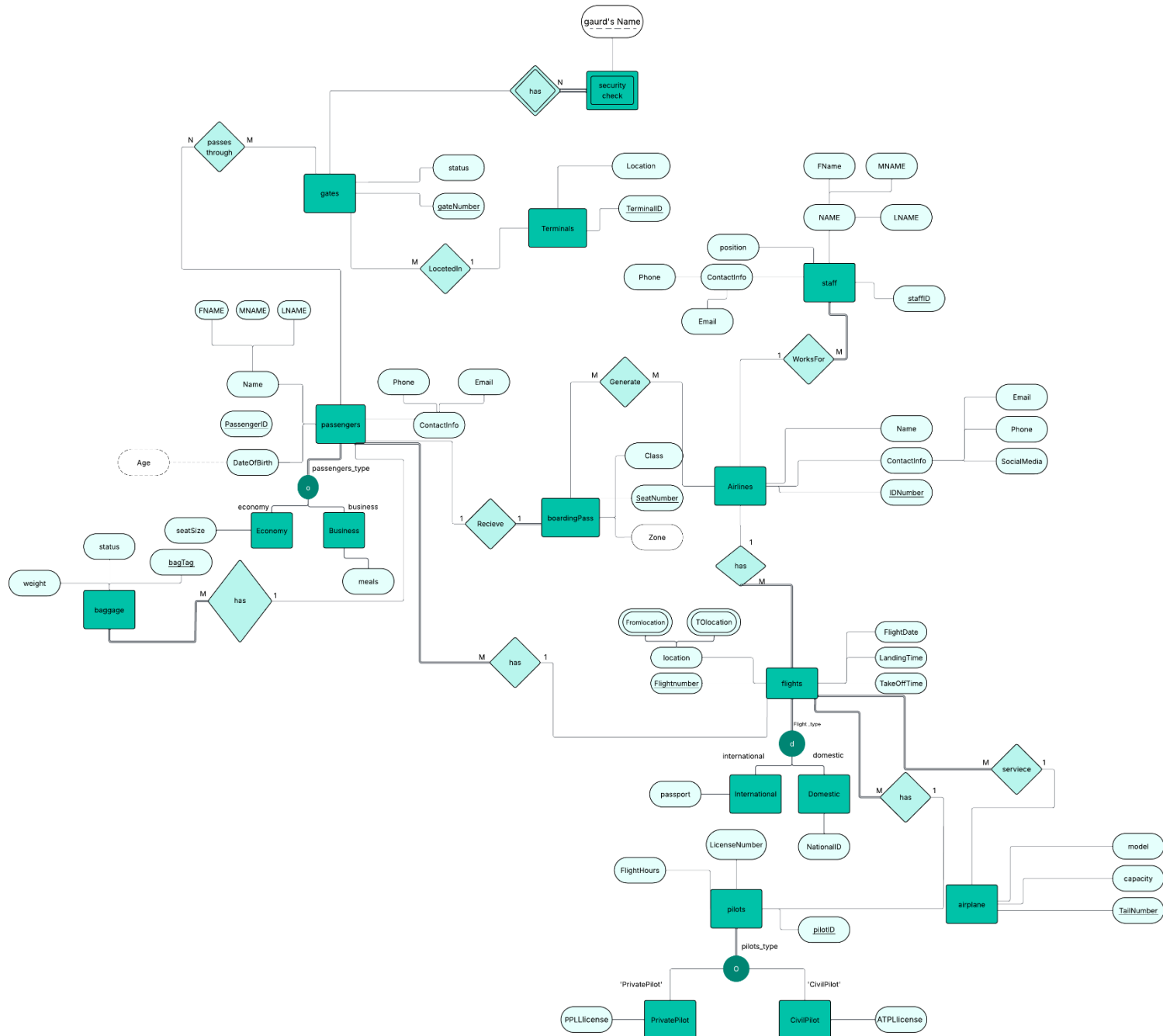
### **1.3 Problem Statement**

- The Airports have problems like flights delays, lost baggage, and slow check-in. These issues happen because the data is not well managed. Using paper or manual work makes it hard to find and update information. The airport need system that stores all data in one place to make the work easier, faster, and more accurate.

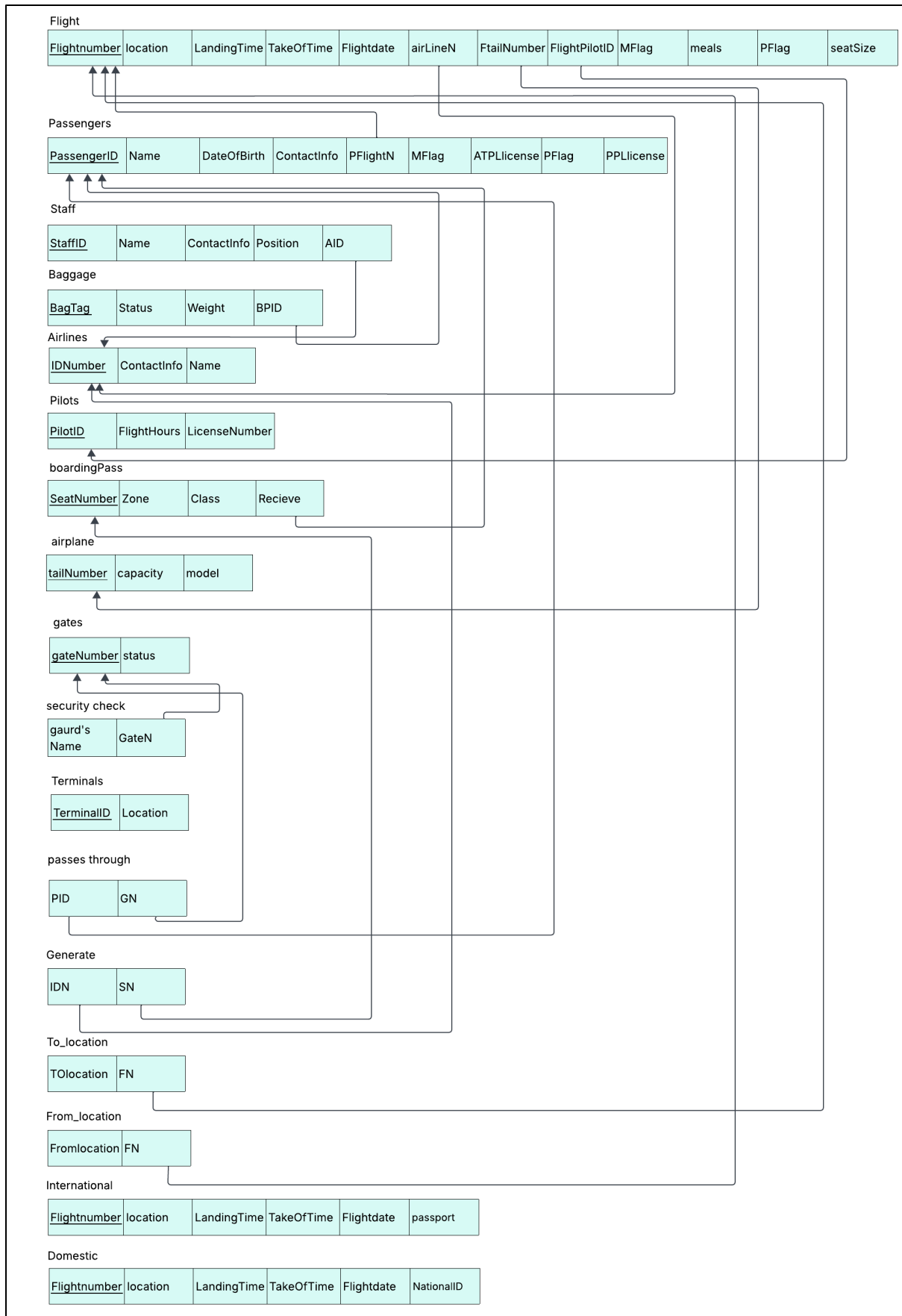
# Chapter 2: Conceptual Model Design

Here are the domain's entities and attributes:

| Entity        | Attributes          |                                                                           |                              |                          |              |
|---------------|---------------------|---------------------------------------------------------------------------|------------------------------|--------------------------|--------------|
| Flights       | <u>FlightNumber</u> | Location<br>({From},{To})                                                 | FlightDate                   | LandingTime              | Takeoff Time |
| passengers    | <u>PassengerID</u>  | Name<br>(FName,MName,LName)                                               | ContactInfo<br>(Phone,Email) | DateOfBirth(Age)         |              |
| Staff         | <u>StaffID</u>      | Name<br>(FName,MName,LName)                                               | Position                     | ContactInfo(Phone,Email) |              |
| Pilots        | <u>PilotID</u>      | LicenseNumber                                                             | FlightHours                  |                          |              |
| Baggage       | <u>BagTag</u>       | Status                                                                    | Weight                       |                          |              |
| Airplane      | <u>TailNumber</u>   | Capacity                                                                  | Model                        |                          |              |
| BoardingPass  | <u>SeatNumber</u>   | Class                                                                     | Zone                         |                          |              |
| Airlines      | <u>IDNumber</u>     | ContactInfo<br>(Phone,Email)                                              | Name                         |                          |              |
| Gates         | <u>GateNumber</u>   | Status                                                                    |                              |                          |              |
| Terminals     | <u>TerminalID</u>   | Location                                                                  |                              |                          |              |
| SecurityCheck | <u>Guard'sName</u>  | It's a week entity depending on the gate entity.                          |                              |                          |              |
| International | Passport            | It's a specialized type of Flight – total participation(dependent)        |                              |                          |              |
| Domestic      | NationalID          | It's a specialized type of Flight - total participation(dependent)        |                              |                          |              |
| privatePilot  | PPL                 | It's a specialized type of Pilot - total participation(overlapping )      |                              |                          |              |
| CivilPilot    | ATPL                | It's a specialized type of Pilot - total participation(overlapping )      |                              |                          |              |
| Economy       | seatSize            | It's a specialized type of passengers - total participation(overlapping ) |                              |                          |              |
| Business      | meals               | It's a specialized type of passengers - total participation(overlapping ) |                              |                          |              |



# Chapter 3: Relational Model Design



# Chapter 4: Database Implementation and SQL Queries

| Table name | Flight                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |              |            |                              |                                 |              |            |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|------------|------------------------------|---------------------------------|--------------|------------|-------|------------|---|-----|----|----|------------------------------|---------------------------------|--------------|---------|---|-----|----|----|------------------------------|---------------------------------|--------------|---------|---|-----|----|----|------------------------------|---------------------------------|--------------|---------|---|-----|----|----|------------------------------|---------------------------------|--------------|---------|---|-----|----|----|------------------------------|---------------------------------|--------------|---------|
| Creating   | <pre>CREATE TABLE Flights(<br/>  FNumber NUMBER(7) CONSTRAINT FLIGHTS_PK PRIMARY KEY,<br/>  FromLocation VARCHAR2(20) CONSTRAINT FLIGHTS_LOC_FROM_NN NOT NULL,<br/>  ToLocation VARCHAR2(20) CONSTRAINT FLIGHTS_LOC_TO_NN NOT NULL,<br/>  LandingTime TIMESTAMP,<br/>  TakeOffTime TIMESTAMP,<br/>  FDate DATE,<br/>  TailNumber VARCHAR2(7),<br/>  CONSTRAINT FLIGHTS_AIRPLANE_FK FOREIGN KEY (TailNumber) REFERENCES AirPlane(TailNumber));<br/>SELECT * FROM Flights;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |              |            |                              |                                 |              |            |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| Inserting  | <pre>INSERT INTO Flights (FNumber, FromLocation, ToLocation, LandingTime, TakeOffTime, FDate, TailNumber)<br/>VALUES (101, 'NY', 'LA', TIMESTAMP '2025-12-25 10:00:00', TIMESTAMP '2025-12-25 08:00:00', DATE '2025-12-25', 'A001JET');<br/>INSERT INTO Flights (FNumber, FromLocation, ToLocation, LandingTime, TakeOffTime, FDate, TailNumber)<br/>VALUES (202, 'LA', 'SF', TIMESTAMP '2025-12-25 12:00:00', TIMESTAMP '2025-12-25 10:30:00', DATE '2025-12-25', 'C003SKY');<br/>INSERT INTO Flights (FNumber, FromLocation, ToLocation, LandingTime, TakeOffTime, FDate, TailNumber)<br/>VALUES (303, 'SF', 'TX', TIMESTAMP '2025-12-26 15:30:00', TIMESTAMP '2025-12-26 12:00:00', DATE '2025-12-26', 'D004MAX');<br/>INSERT INTO Flights (FNumber, FromLocation, ToLocation, LandingTime, TakeOffTime, FDate, TailNumber)<br/>VALUES (404, 'TX', 'MI', TIMESTAMP '2025-12-26 20:00:00', TIMESTAMP '2025-12-26 17:00:00', DATE '2025-12-26', 'A001JET');<br/>INSERT INTO Flights (FNumber, FromLocation, ToLocation, LandingTime, TakeOffTime, FDate, TailNumber)<br/>VALUES (505, 'MI', 'NY', TIMESTAMP '2025-12-27 09:00:00', TIMESTAMP '2025-12-27 07:00:00', DATE '2025-12-27', 'B002PRO');</pre>      |              |            |                              |                                 |              |            |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| Data       | <div><div>Script Output xQuery Result x</div><div><div><div></div><div></div><div></div><div></div></div><div>SQL   All Rows Fetched: 5 in 0.007 seconds</div></div><table><tr><th></th><th>FNUMBER</th><th>FROMLOCATION</th><th>TOLOCATION</th><th>LANDINGTIME</th><th>TAKEOFTIME</th><th>FDATE</th><th>TAILNUMBER</th></tr><tr><td>1</td><td>101</td><td>NY</td><td>LA</td><td>25-DEC-25 10.00.00.000000000</td><td>AM 25-DEC-25 08.00.00.000000000</td><td>AM 25-DEC-25</td><td>A001JET</td></tr><tr><td>2</td><td>303</td><td>SF</td><td>TX</td><td>26-DEC-25 03.30.00.000000000</td><td>PM 26-DEC-25 12.00.00.000000000</td><td>PM 26-DEC-25</td><td>D004MAX</td></tr><tr><td>3</td><td>404</td><td>TX</td><td>MI</td><td>26-DEC-25 08.00.00.000000000</td><td>PM 26-DEC-25 05.00.00.000000000</td><td>PM 26-DEC-25</td><td>A001JET</td></tr><tr><td>4</td><td>505</td><td>MI</td><td>NY</td><td>27-DEC-25 09.00.00.000000000</td><td>AM 27-DEC-25 07.00.00.000000000</td><td>AM 27-DEC-25</td><td>B002PRO</td></tr><tr><td>5</td><td>202</td><td>LA</td><td>SF</td><td>25-DEC-25 12.00.00.000000000</td><td>PM 25-DEC-25 10.30.00.000000000</td><td>AM 25-DEC-25</td><td>C003SKY</td></tr></table></div> |              | FNUMBER    | FROMLOCATION                 | TOLOCATION                      | LANDINGTIME  | TAKEOFTIME | FDATE | TAILNUMBER | 1 | 101 | NY | LA | 25-DEC-25 10.00.00.000000000 | AM 25-DEC-25 08.00.00.000000000 | AM 25-DEC-25 | A001JET | 2 | 303 | SF | TX | 26-DEC-25 03.30.00.000000000 | PM 26-DEC-25 12.00.00.000000000 | PM 26-DEC-25 | D004MAX | 3 | 404 | TX | MI | 26-DEC-25 08.00.00.000000000 | PM 26-DEC-25 05.00.00.000000000 | PM 26-DEC-25 | A001JET | 4 | 505 | MI | NY | 27-DEC-25 09.00.00.000000000 | AM 27-DEC-25 07.00.00.000000000 | AM 27-DEC-25 | B002PRO | 5 | 202 | LA | SF | 25-DEC-25 12.00.00.000000000 | PM 25-DEC-25 10.30.00.000000000 | AM 25-DEC-25 | C003SKY |
|            | FNUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | FROMLOCATION | TOLOCATION | LANDINGTIME                  | TAKEOFTIME                      | FDATE        | TAILNUMBER |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| 1          | 101                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | NY           | LA         | 25-DEC-25 10.00.00.000000000 | AM 25-DEC-25 08.00.00.000000000 | AM 25-DEC-25 | A001JET    |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| 2          | 303                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | SF           | TX         | 26-DEC-25 03.30.00.000000000 | PM 26-DEC-25 12.00.00.000000000 | PM 26-DEC-25 | D004MAX    |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| 3          | 404                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | TX           | MI         | 26-DEC-25 08.00.00.000000000 | PM 26-DEC-25 05.00.00.000000000 | PM 26-DEC-25 | A001JET    |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| 4          | 505                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | MI           | NY         | 27-DEC-25 09.00.00.000000000 | AM 27-DEC-25 07.00.00.000000000 | AM 27-DEC-25 | B002PRO    |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |
| 5          | 202                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | LA           | SF         | 25-DEC-25 12.00.00.000000000 | PM 25-DEC-25 10.30.00.000000000 | AM 25-DEC-25 | C003SKY    |       |            |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |   |     |    |    |                              |                                 |              |         |



| Table name | Staff                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                     |          |              |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|---------------------|----------|--------------|-------|----------|---|------|----------|--------------------|---------|--------------|---|------|------------|--------------------|---------|-------------|---|------|-----------|---------------------|---------|----------|---|------|------------|---------------------|---------|------------|---|------|-----------|-------------------|---------|------------|
| Creating   | <pre>CREATE TABLE Staff(StaffID NUMBER(7) CONSTRAINT PK3 PRIMARY KEY, FullName VARCHAR2(50) CONSTRAINT NN4 NOT NULL, Email VARCHAR2(20) CONSTRAINT Staff_EMAIL_NN1 NOT NULL CONSTRAINT Staff_EMAIL_UC1 UNIQUE, Phone VARCHAR2(20) CONSTRAINT Staff_PHONE_NN2 NOT NULL CONSTRAINT Staff_PHONE_UC2 UNIQUE, Position VARCHAR2(20) CONSTRAINT NN6 NOT NULL); SELECT * FROM Staff;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                     |          |              |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| Inserting  | <pre>INSERT INTO Staff (StaffID, FullName, Email, Phone, Position) VALUES (1001, 'Jane Doe', 'jane.d@airport.com', '5551234', 'Ground Staff'); INSERT INTO Staff (StaffID, FullName, Email, Phone, Position) VALUES (1002, 'Alex Smith', 'alex.s@airport.com', '5554321', 'Maintenance'); INSERT INTO Staff (StaffID, FullName, Email, Phone, Position) VALUES (1003, 'Chris Lee', 'chris.l@airport.com', '5556789', 'Security'); INSERT INTO Staff (StaffID, FullName, Email, Phone, Position) VALUES (1004, 'Patty Ryan', 'patty.r@airport.com', '5559876', 'Gate Agent'); INSERT INTO Staff (StaffID, FullName, Email, Phone, Position) VALUES (1005, 'Sam Brown', 'sam.b@airport.com', '5550000', 'Supervisor');</pre>                                                                                                       |            |                     |          |              |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| Data       | <div><div>Script Output x</div><div>Query Result x</div><div><div>SQL   All Rows Fetched: 5 in 0.006 seconds</div><table><tr><th></th><th>STAFFID</th><th>FULLNAME</th><th>EMAIL</th><th>PHONE</th><th>POSITION</th></tr><tr><td>1</td><td>1001</td><td>Jane Doe</td><td>jane.d@airport.com</td><td>5551234</td><td>Ground Staff</td></tr><tr><td>2</td><td>1002</td><td>Alex Smith</td><td>alex.s@airport.com</td><td>5554321</td><td>Maintenance</td></tr><tr><td>3</td><td>1003</td><td>Chris Lee</td><td>chris.l@airport.com</td><td>5556789</td><td>Security</td></tr><tr><td>4</td><td>1004</td><td>Patty Ryan</td><td>patty.r@airport.com</td><td>5559876</td><td>Gate Agent</td></tr><tr><td>5</td><td>1005</td><td>Sam Brown</td><td>sam.b@airport.com</td><td>5550000</td><td>Supervisor</td></tr></table></div></div> |            | STAFFID             | FULLNAME | EMAIL        | PHONE | POSITION | 1 | 1001 | Jane Doe | jane.d@airport.com | 5551234 | Ground Staff | 2 | 1002 | Alex Smith | alex.s@airport.com | 5554321 | Maintenance | 3 | 1003 | Chris Lee | chris.l@airport.com | 5556789 | Security | 4 | 1004 | Patty Ryan | patty.r@airport.com | 5559876 | Gate Agent | 5 | 1005 | Sam Brown | sam.b@airport.com | 5550000 | Supervisor |
|            | STAFFID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | FULLNAME   | EMAIL               | PHONE    | POSITION     |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| 1          | 1001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Jane Doe   | jane.d@airport.com  | 5551234  | Ground Staff |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| 2          | 1002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Alex Smith | alex.s@airport.com  | 5554321  | Maintenance  |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| 3          | 1003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Chris Lee  | chris.l@airport.com | 5556789  | Security     |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| 4          | 1004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Patty Ryan | patty.r@airport.com | 5559876  | Gate Agent   |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |
| 5          | 1005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Sam Brown  | sam.b@airport.com   | 5550000  | Supervisor   |       |          |   |      |          |                    |         |              |   |      |            |                    |         |             |   |      |           |                     |         |          |   |      |            |                     |         |            |   |      |           |                   |         |            |

| Table name | Passengers                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |               |             |                  |             |         |       |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|-------------|------------------|-------------|---------|-------|---------|-------|---|---------|---------------|-----------|----------------|---------|-----|----|---|---------|--------------|-----------|--------------|---------|-----|----|---|---------|--------------|-----------|----------------|---------|-----|----|---|---------|------------|-----------|----------------|---------|-----|----|---|---------|---------------|-----------|------------------|---------|-----|---|
| Creating   | <pre>CREATE TABLE Passengers(PID VARCHAR2(7) CONSTRAINT PASSENGERS_PK PRIMARY KEY, FullName VARCHAR2(50) CONSTRAINT PASSENGERS_NAME_NN NOT NULL, DateOfBirth DATE, Email VARCHAR2(20) CONSTRAINT PASSENGERS_EMAIL_NN NOT NULL CONSTRAINT PASSENGERS_EMAIL_UC UNIQUE, Phone VARCHAR2(20) CONSTRAINT PASSENGERS_PHONE_NN NOT NULL CONSTRAINT PASSENGERS_PHONE_UC UNIQUE, FNumber NUMBER(7),GateN Number(2), CONSTRAINT PASSENGERS_FLIGHT_FK FOREIGN KEY (FNumber) REFERENCES Flights(FNumber), CONSTRAINT PASSENGERS_GATEN_FK FOREIGN KEY (GateN) REFERENCES Gates(GateN)); SELECT * FROM Passengers;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                            |               |             |                  |             |         |       |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| Inserting  | <pre>INSERT INTO Passengers (PID, FullName, DateOfBirth, Email, Phone, FNumber, GateN) VALUES ('P000001', 'Alice Johnson', DATE '1985-01-15', 'alice@mail.com', '5551111', 101, 11); INSERT INTO Passengers (PID, FullName, DateOfBirth, Email, Phone, FNumber, GateN) VALUES ('P000002', 'Bob Williams', DATE '1990-05-20', 'bob@mail.com', '5552222', 101, 11); INSERT INTO Passengers (PID, FullName, DateOfBirth, Email, Phone, FNumber, GateN) VALUES ('P000003', 'Charlie Brown', DATE '2000-11-30', 'charlie@mail.com', '5553333', 202, 2); INSERT INTO Passengers (PID, FullName, DateOfBirth, Email, Phone, FNumber, GateN) VALUES ('P000004', 'Diana Prince', DATE '1975-03-01', 'diana@mail.com', '5554444', 303, 10); INSERT INTO Passengers (PID, FullName, DateOfBirth, Email, Phone, FNumber, GateN) VALUES ('P000005', 'Ethan Hunt', DATE '1965-08-25', 'ethan@mail.com', '5555555', 505, 20);</pre>                                                                                                                                 |               |             |                  |             |         |       |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| Data       | <div><div>Script Output x</div><div>Query Result x</div><div> All Rows Fetched: 5 in 0.003 seconds</div><table><tr><th></th><th>PID</th><th>FULLNAME</th><th>DATEOFBIRTH</th><th>EMAIL</th><th>PHONE</th><th>FNUMBER</th><th>GATEN</th></tr><tr><td>1</td><td>P000001</td><td>Alice Johnson</td><td>15-JAN-85</td><td>alice@mail.com</td><td>5551111</td><td>101</td><td>11</td></tr><tr><td>2</td><td>P000002</td><td>Bob Williams</td><td>20-MAY-90</td><td>bob@mail.com</td><td>5552222</td><td>101</td><td>11</td></tr><tr><td>3</td><td>P000004</td><td>Diana Prince</td><td>01-MAR-75</td><td>diana@mail.com</td><td>5554444</td><td>303</td><td>10</td></tr><tr><td>4</td><td>P000005</td><td>Ethan Hunt</td><td>25-AUG-65</td><td>ethan@mail.com</td><td>5555555</td><td>505</td><td>20</td></tr><tr><td>5</td><td>P000003</td><td>Charlie Brown</td><td>30-NOV-00</td><td>charlie@mail.com</td><td>5553333</td><td>202</td><td>2</td></tr></table></div> |               | PID         | FULLNAME         | DATEOFBIRTH | EMAIL   | PHONE | FNUMBER | GATEN | 1 | P000001 | Alice Johnson | 15-JAN-85 | alice@mail.com | 5551111 | 101 | 11 | 2 | P000002 | Bob Williams | 20-MAY-90 | bob@mail.com | 5552222 | 101 | 11 | 3 | P000004 | Diana Prince | 01-MAR-75 | diana@mail.com | 5554444 | 303 | 10 | 4 | P000005 | Ethan Hunt | 25-AUG-65 | ethan@mail.com | 5555555 | 505 | 20 | 5 | P000003 | Charlie Brown | 30-NOV-00 | charlie@mail.com | 5553333 | 202 | 2 |
|            | PID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | FULLNAME      | DATEOFBIRTH | EMAIL            | PHONE       | FNUMBER | GATEN |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| 1          | P000001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Alice Johnson | 15-JAN-85   | alice@mail.com   | 5551111     | 101     | 11    |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| 2          | P000002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Bob Williams  | 20-MAY-90   | bob@mail.com     | 5552222     | 101     | 11    |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| 3          | P000004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Diana Prince  | 01-MAR-75   | diana@mail.com   | 5554444     | 303     | 10    |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| 4          | P000005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Ethan Hunt    | 25-AUG-65   | ethan@mail.com   | 5555555     | 505     | 20    |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |
| 5          | P000003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Charlie Brown | 30-NOV-00   | charlie@mail.com | 5553333     | 202     | 2     |         |       |   |         |               |           |                |         |     |    |   |         |              |           |              |         |     |    |   |         |              |           |                |         |     |    |   |         |            |           |                |         |     |    |   |         |               |           |                  |         |     |   |

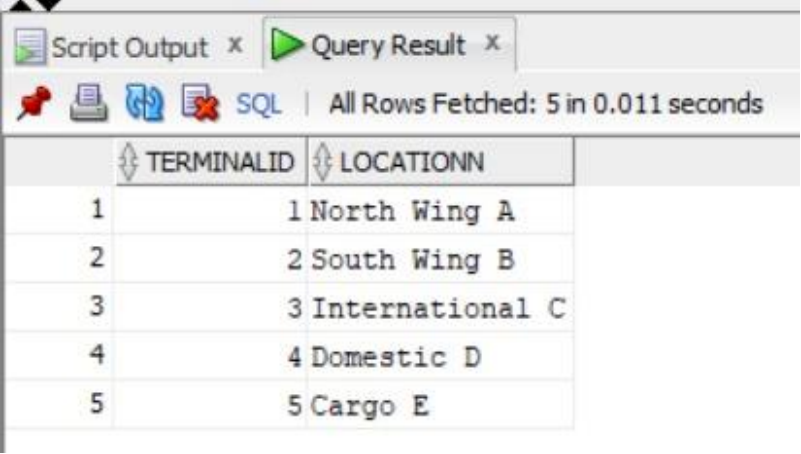
| Table name | Pilot                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |             |              |             |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|--------------|-------------|--------------|---------|---|------|------|--------|-----|---|------|------|--------|-----|---|------|-------|--------|-----|---|------|------|--------|-----|---|------|-----|--------|-----|
| Creating   | <pre>CREATE TABLE Pilots(PilotID NUMBER(7) CONSTRAINT PK6 PRIMARY KEY,<br/>FlightHours NUMBER(6) CONSTRAINT NN9 NOT NULL,<br/>LicensNumber VARCHAR2(20) CONSTRAINT NN10 NOT NULL,<br/>FNumber NUMBER(7),CONSTRAINT FK7 FOREIGN KEY (FNumber) REFERENCES Flights(FNumber));<br/>SELECT * FROM Pilots;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |             |              |             |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| Inserting  | <pre>INSERT INTO Pilots (PilotID, FlightHours, LicensNumber, FNumber) VALUES (7001, 8500, 'LNC001', 101);<br/>INSERT INTO Pilots (PilotID, FlightHours, LicensNumber, FNumber) VALUES (7002, 5200, 'LNC002', 202);<br/>INSERT INTO Pilots (PilotID, FlightHours, LicensNumber, FNumber) VALUES (7003, 12000, 'LNC003', 303);<br/>INSERT INTO Pilots (PilotID, FlightHours, LicensNumber, FNumber) VALUES (7004, 3100, 'LNC004', 404);<br/>INSERT INTO Pilots (PilotID, FlightHours, LicensNumber, FNumber) VALUES (7005, 950, 'LNC005', 505);</pre>                                                                                                                                                                                                                                                                                                                                                                                        |             |              |             |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| Data       | <div><div>Script Output xQuery Result x</div><div>    SQL   All Rows Fetched: 5 in 0.005 seconds</div><table><tr><th></th><th>PILOTID</th><th>FLIGHTHOURS</th><th>LICENSNUMBER</th><th>FNUMBER</th></tr><tr><td>1</td><td>7001</td><td>8500</td><td>LNC001</td><td>101</td></tr><tr><td>2</td><td>7002</td><td>5200</td><td>LNC002</td><td>202</td></tr><tr><td>3</td><td>7003</td><td>12000</td><td>LNC003</td><td>303</td></tr><tr><td>4</td><td>7004</td><td>3100</td><td>LNC004</td><td>404</td></tr><tr><td>5</td><td>7005</td><td>950</td><td>LNC005</td><td>505</td></tr></table></div> |             | PILOTID      | FLIGHTHOURS | LICENSNUMBER | FNUMBER | 1 | 7001 | 8500 | LNC001 | 101 | 2 | 7002 | 5200 | LNC002 | 202 | 3 | 7003 | 12000 | LNC003 | 303 | 4 | 7004 | 3100 | LNC004 | 404 | 5 | 7005 | 950 | LNC005 | 505 |
|            | PILOTID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | FLIGHTHOURS | LICENSNUMBER | FNUMBER     |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| 1          | 7001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 8500        | LNC001       | 101         |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| 2          | 7002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5200        | LNC002       | 202         |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| 3          | 7003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12000       | LNC003       | 303         |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| 4          | 7004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 3100        | LNC004       | 404         |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |
| 5          | 7005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 950         | LNC005       | 505         |              |         |   |      |      |        |     |   |      |      |        |     |   |      |       |        |     |   |      |      |        |     |   |      |     |        |     |



| Table name | Airline                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |             |               |             |              |         |             |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|---------------|-------------|--------------|---------|-------------|---------|---------|---|-----|---------|-------------|---------|----------|-----|------|---|-----|---------|---------------|---------|-------------|-----|------|---|-----|----------|--------------|---------|--------------|-----|------|---|-----|---------|--------------|---------|-------------|-----|------|---|-----|----------|---------------|---------|--------------|-----|------|
| Creating   | <pre>CREATE TABLE AirLines(IDNumber NUMBER(7) CONSTRAINT PK5 PRIMARY KEY, AirlineName VARCHAR2(20) CONSTRAINT U3 UNIQUE, Email VARCHAR2(20) CONSTRAINT AirLines_EMAIL_NN3 NOT NULL CONSTRAINT AirLines_EMAIL_UC3 UNIQUE, Phone VARCHAR2(20) CONSTRAINT AirLines_PHONE_NN4 NOT NULL CONSTRAINT AirLines_PHONE_UC4 UNIQUE, SocialMedia VARCHAR2(20) CONSTRAINT AirLines_SocialMedia_NN4 NOT NULL CONSTRAINT AirLines_SocialMedia_UC4 UNIQUE, FNumber NUMBER(7),StaffID NUMBER(7), CONSTRAINT FK5 FOREIGN KEY (FNumber) REFERENCES Flights(FNumber), CONSTRAINT FK6 FOREIGN KEY (StaffID) REFERENCES Staff(StaffID)); SELECT * FROM AirLines;</pre>                                                                                                                                                                                                                                                                                                 |             |               |             |              |         |             |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| Inserting  | <pre>INSERT INTO AirLines (IDNumber, AirlineName, Email, Phone, SocialMedia, FNumber, StaffID) VALUES (501, 'SkyHigh', 'sky@fly.com', '8001111', '@SkyHigh', 101, 1001); INSERT INTO AirLines (IDNumber, AirlineName, Email, Phone, SocialMedia, FNumber, StaffID) VALUES (502, 'Oceanic', 'ocean@fly.com', '8002222', '@OceanicAir', 202, 1002); INSERT INTO AirLines (IDNumber, AirlineName, Email, Phone, SocialMedia, FNumber, StaffID) VALUES (503, 'StarWing', 'star@fly.com', '8003333', '@StarWingFly', 303, 1003); INSERT INTO AirLines (IDNumber, AirlineName, Email, Phone, SocialMedia, FNumber, StaffID) VALUES (504, 'FastFly', 'fast@fly.com', '8004444', '@FastFlyNow', 404, 1004); INSERT INTO AirLines (IDNumber, AirlineName, Email, Phone, SocialMedia, FNumber, StaffID) VALUES (505, 'GlobeAir', 'globe@fly.com', '8005555', '@GlobeAirInt', 505, 1005);</pre>                                                             |             |               |             |              |         |             |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| Data       | <div><div>Script Output x</div><div>Query Result x</div><div>SQL   All Rows Fetched: 5 in 0.005 seconds</div><table><tr><th></th><th>IDNUMBER</th><th>AIRLINENAME</th><th>EMAIL</th><th>PHONE</th><th>SOCIALMEDIA</th><th>FNUMBER</th><th>STAFFID</th></tr><tr><td>1</td><td>501</td><td>SkyHigh</td><td>sky@fly.com</td><td>8001111</td><td>@SkyHigh</td><td>101</td><td>1001</td></tr><tr><td>2</td><td>502</td><td>Oceanic</td><td>ocean@fly.com</td><td>8002222</td><td>@OceanicAir</td><td>202</td><td>1002</td></tr><tr><td>3</td><td>503</td><td>StarWing</td><td>star@fly.com</td><td>8003333</td><td>@StarWingFly</td><td>303</td><td>1003</td></tr><tr><td>4</td><td>504</td><td>FastFly</td><td>fast@fly.com</td><td>8004444</td><td>@FastFlyNow</td><td>404</td><td>1004</td></tr><tr><td>5</td><td>505</td><td>GlobeAir</td><td>globe@fly.com</td><td>8005555</td><td>@GlobeAirInt</td><td>505</td><td>1005</td></tr></table></div> |             | IDNUMBER      | AIRLINENAME | EMAIL        | PHONE   | SOCIALMEDIA | FNUMBER | STAFFID | 1 | 501 | SkyHigh | sky@fly.com | 8001111 | @SkyHigh | 101 | 1001 | 2 | 502 | Oceanic | ocean@fly.com | 8002222 | @OceanicAir | 202 | 1002 | 3 | 503 | StarWing | star@fly.com | 8003333 | @StarWingFly | 303 | 1003 | 4 | 504 | FastFly | fast@fly.com | 8004444 | @FastFlyNow | 404 | 1004 | 5 | 505 | GlobeAir | globe@fly.com | 8005555 | @GlobeAirInt | 505 | 1005 |
|            | IDNUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | AIRLINENAME | EMAIL         | PHONE       | SOCIALMEDIA  | FNUMBER | STAFFID     |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| 1          | 501                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | SkyHigh     | sky@fly.com   | 8001111     | @SkyHigh     | 101     | 1001        |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| 2          | 502                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Oceanic     | ocean@fly.com | 8002222     | @OceanicAir  | 202     | 1002        |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| 3          | 503                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | StarWing    | star@fly.com  | 8003333     | @StarWingFly | 303     | 1003        |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| 4          | 504                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | FastFly     | fast@fly.com  | 8004444     | @FastFlyNow  | 404     | 1004        |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |
| 5          | 505                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | GlobeAir    | globe@fly.com | 8005555     | @GlobeAirInt | 505     | 1005        |         |         |   |     |         |             |         |          |     |      |   |     |         |               |         |             |     |      |   |     |          |              |         |              |     |      |   |     |         |              |         |             |     |      |   |     |          |               |         |              |     |      |

| Table name | Airplane                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |          |       |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|--|------------|----------|-------|---|---------|-----|-----|---|---------|-----|------|---|---------|-----|-----|---|---------|----|-----|---|---------|-----|------|
| Creating   | <pre>CREATE TABLE AirPlane(TailNumber VARCHAR2(7) CONSTRAINT PK8 PRIMARY KEY, Capacity NUMBER(5) CONSTRAINT NN11 NOT NULL, Model VARCHAR2(10) );<br/>SELECT * FROM AirPlane;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |          |       |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| Inserting  | <pre>INSERT INTO AirPlane (TailNumber, Capacity, Model) VALUES ('A001JET', 180, '737');<br/>INSERT INTO AirPlane (TailNumber, Capacity, Model) VALUES ('B002PRO', 350, 'A380');<br/>INSERT INTO AirPlane (TailNumber, Capacity, Model) VALUES ('C003SKY', 120, 'E190');<br/>INSERT INTO AirPlane (TailNumber, Capacity, Model) VALUES ('D004MAX', 250, '787');<br/>INSERT INTO AirPlane (TailNumber, Capacity, Model) VALUES ('E005GLD', 80, 'CRJ');</pre>                                                                                                                                                                                                                                                                                                                                                                                                     |          |       |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| Data       | <div><div>Script Output x</div><div>Query Result x</div><div>    SQL   All Rows Fetched: 5 in 0.007 seconds</div><table><tr><th></th><th>TAILNUMBER</th><th>CAPACITY</th><th>MODEL</th></tr><tr><td>1</td><td>A001JET</td><td>180</td><td>737</td></tr><tr><td>2</td><td>B002PRO</td><td>350</td><td>A380</td></tr><tr><td>3</td><td>D004MAX</td><td>250</td><td>787</td></tr><tr><td>4</td><td>E005GLD</td><td>80</td><td>CRJ</td></tr><tr><td>5</td><td>C003SKY</td><td>120</td><td>E190</td></tr></table></div> |          |       |  | TAILNUMBER | CAPACITY | MODEL | 1 | A001JET | 180 | 737 | 2 | B002PRO | 350 | A380 | 3 | D004MAX | 250 | 787 | 4 | E005GLD | 80 | CRJ | 5 | C003SKY | 120 | E190 |
|            | TAILNUMBER                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | CAPACITY | MODEL |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| 1          | A001JET                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 180      | 737   |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| 2          | B002PRO                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 350      | A380  |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| 3          | D004MAX                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 250      | 787   |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| 4          | E005GLD                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 80       | CRJ   |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |
| 5          | C003SKY                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 120      | E190  |  |            |          |       |   |         |     |     |   |         |     |      |   |         |     |     |   |         |    |     |   |         |     |      |

| Table name | Gate                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |          |            |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
|------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|------------|---------|------------|---|---|------|---|---|---|--------|---|---|----|------|---|---|----|----------|---|---|----|------|---|
| Creating   | <pre>CREATE TABLE Gates (GateN Number(2) CONSTRAINT PK9 PRIMARY KEY, Statuss VARCHAR (20) NOT NULL,TerminalID NUMBER(7), CONSTRAINT FK10 FOREIGN KEY (TerminalID) REFERENCES Terminals (TerminalID)); SELECT * FROM Gates;</pre>                                                                                                                                                                                                                                                                                                                        |          |            |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| Inserting  | <pre>INSERT INTO Gates (GateN, Statuss, TerminalID) VALUES (1, 'Open', 1); INSERT INTO Gates (GateN, Statuss, TerminalID) VALUES (2, 'Closed', 1); INSERT INTO Gates (GateN, Statuss, TerminalID) VALUES (10, 'Open', 2); INSERT INTO Gates (GateN, Statuss, TerminalID) VALUES (11, 'Boarding', 3); INSERT INTO Gates (GateN, Statuss, TerminalID) VALUES (20, 'Open', 4);</pre>                                                                                                                                                                       |          |            |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| Data       | <div><div>Script Output xQuery Result x</div><div>   All Rows Fetched: 5 in 0.005 seconds</div><table><tr><th></th><th>GATEN</th><th>STATUSS</th><th>TERMINALID</th></tr><tr><td>1</td><td>1</td><td>Open</td><td>1</td></tr><tr><td>2</td><td>2</td><td>Closed</td><td>1</td></tr><tr><td>3</td><td>10</td><td>Open</td><td>2</td></tr><tr><td>4</td><td>11</td><td>Boarding</td><td>3</td></tr><tr><td>5</td><td>20</td><td>Open</td><td>4</td></tr></table></div> |          | GATEN      | STATUSS | TERMINALID | 1 | 1 | Open | 1 | 2 | 2 | Closed | 1 | 3 | 10 | Open | 2 | 4 | 11 | Boarding | 3 | 5 | 20 | Open | 4 |
|            | GATEN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | STATUSS  | TERMINALID |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| 1          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Open     | 1          |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| 2          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Closed   | 1          |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| 3          | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Open     | 2          |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| 4          | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Boarding | 3          |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |
| 5          | 20                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | Open     | 4          |         |            |   |   |      |   |   |   |        |   |   |    |      |   |   |    |          |   |   |    |      |   |

| Table name | Terminal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                 |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
|------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|--|------------|------------|---|---|--------------|---|---|--------------|---|---|-----------------|---|---|------------|---|---|---------|
| Creating   | <pre>CREATE TABLE Terminals(TerminalID NUMBER(2) CONSTRAINT PK10 PRIMARY KEY,<br/>Locationnn VARCHAR (20));<br/>SELECT * FROM Terminals;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                 |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| Inserting  | <pre>INSERT INTO Terminals (TerminalID, Locationnn) VALUES (1, 'North Wing A');<br/>INSERT INTO Terminals (TerminalID, Locationnn) VALUES (2, 'South Wing B');<br/>INSERT INTO Terminals (TerminalID, Locationnn) VALUES (3, 'International C');<br/>INSERT INTO Terminals (TerminalID, Locationnn) VALUES (4, 'Domestic D');<br/>INSERT INTO Terminals (TerminalID, Locationnn) VALUES (5, 'Cargo E');</pre>                                                                                                                                                                                                                                                                                                                                                                      |                 |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| Data       |  <p>The screenshot shows a database query result window with two tabs: 'Script Output' and 'Query Result'. The 'Query Result' tab is active, displaying a table with two columns: 'TERMINALID' and 'LOCATIONNN'. The table contains five rows of data, corresponding to the insert statements in the previous step.</p> <table border="1"><thead><tr><th></th><th>TERMINALID</th><th>LOCATIONNN</th></tr></thead><tbody><tr><td>1</td><td>1</td><td>North Wing A</td></tr><tr><td>2</td><td>2</td><td>South Wing B</td></tr><tr><td>3</td><td>3</td><td>International C</td></tr><tr><td>4</td><td>4</td><td>Domestic D</td></tr><tr><td>5</td><td>5</td><td>Cargo E</td></tr></tbody></table> |                 |  | TERMINALID | LOCATIONNN | 1 | 1 | North Wing A | 2 | 2 | South Wing B | 3 | 3 | International C | 4 | 4 | Domestic D | 5 | 5 | Cargo E |
|            | TERMINALID                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | LOCATIONNN      |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| 1          | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | North Wing A    |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| 2          | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | South Wing B    |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| 3          | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | International C |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| 4          | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Domestic D      |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |
| 5          | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Cargo E         |  |            |            |   |   |              |   |   |              |   |   |                 |   |   |            |   |   |         |



| Table name | Baggage                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |        |            |         |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------|------------|---------|--------|-----|---|---------|------|------------|---------|---|---------|----|------------|---------|---|---------|----|------------|---------|---|---------|-----|----------|---------|---|---------|------|---------|---------|
| Creating   | <pre>CREATE TABLE Baggage (BagTag VARCHAR2(10) CONSTRAINT PK4 PRIMARY KEY, Weight NUMBER(6,2) CONSTRAINT Weight_C CHECK(Weight &gt; 0), Status VARCHAR2(20) CONSTRAINT NN7 NOT NULL, PID VARCHAR2(7), CONSTRAINT FK4 FOREIGN KEY (PID) REFERENCES Passengers(PID)); SELECT * FROM baggage;</pre>                                                                                                                                                                                                                              |        |            |         |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| Inserting  | <pre>INSERT INTO Baggage (BagTag, Weight, Status, PID) VALUES ('BTA0001', 15.5, 'Checked-In', 'P000001'); INSERT INTO Baggage (BagTag, Weight, Status, PID) VALUES ('BTA0002', 22.0, 'In-Transit', 'P000001'); INSERT INTO Baggage (BagTag, Weight, Status, PID) VALUES ('BTB0003', 18.0, 'Checked-In', 'P000002'); INSERT INTO Baggage (BagTag, Weight, Status, PID) VALUES ('BTC0004', 5.3, 'Carry-On', 'P000003'); INSERT INTO Baggage (BagTag, Weight, Status, PID) VALUES ('BTD0005', 28.9, 'Missing', 'P000004');</pre> |        |            |         |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| Data       | <table><tr><th></th><th>BAGTAG</th><th>WEIGHT</th><th>STATUS</th><th>PID</th></tr><tr><td>1</td><td>BTA0001</td><td>15.5</td><td>Checked-In</td><td>P000001</td></tr><tr><td>2</td><td>BTA0002</td><td>22</td><td>In-Transit</td><td>P000001</td></tr><tr><td>3</td><td>BTB0003</td><td>18</td><td>Checked-In</td><td>P000002</td></tr><tr><td>4</td><td>BTC0004</td><td>5.3</td><td>Carry-On</td><td>P000003</td></tr><tr><td>5</td><td>BTD0005</td><td>28.9</td><td>Missing</td><td>P000004</td></tr></table>               |        | BAGTAG     | WEIGHT  | STATUS | PID | 1 | BTA0001 | 15.5 | Checked-In | P000001 | 2 | BTA0002 | 22 | In-Transit | P000001 | 3 | BTB0003 | 18 | Checked-In | P000002 | 4 | BTC0004 | 5.3 | Carry-On | P000003 | 5 | BTD0005 | 28.9 | Missing | P000004 |
|            | BAGTAG                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | WEIGHT | STATUS     | PID     |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| 1          | BTA0001                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 15.5   | Checked-In | P000001 |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| 2          | BTA0002                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 22     | In-Transit | P000001 |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| 3          | BTB0003                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 18     | Checked-In | P000002 |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| 4          | BTC0004                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 5.3    | Carry-On   | P000003 |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |
| 5          | BTD0005                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 28.9   | Missing    | P000004 |        |     |   |         |      |            |         |   |         |    |            |         |   |         |    |            |         |   |         |     |          |         |   |         |      |         |         |

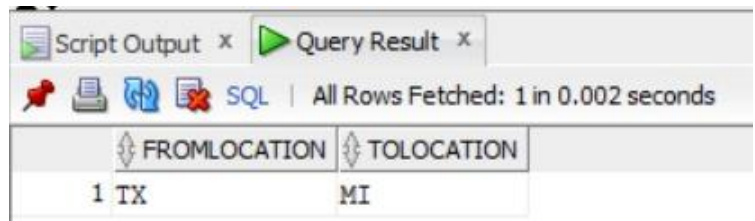


| Table name | BoardingPass                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |      |          |         |          |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
|------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|----------|---------|----------|--|-------|------|-------|-----|----------|---|----|---|-------|---------|-----|---|----|---|----------|---------|-----|---|----|---|-------|---------|-----|---|----|---|---------|---------|-----|---|----|---|---------|---------|-----|
| Creating   | <pre>CREATE TABLE BoardingPass(SeatN VARCHAR2(2) CONSTRAINT PK7 PRIMARY KEY, Zone NUMBER(2),Class VARCHAR2(15), PID VARCHAR2(7),IDNumber NUMBER(7), CONSTRAINT FK8 FOREIGN KEY (PID) REFERENCES Passengers(PID), CONSTRAINT FK9 FOREIGN KEY (IDNumber) REFERENCES AirLines(IDNumber)); SELECT * FROM BoardingPass;</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |          |         |          |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| Inserting  | <pre>INSERT INTO BoardingPass (SeatN, Zone, Class, PID, IDNumber) VALUES ('2A', 1, 'First', 'P000001', 501); INSERT INTO BoardingPass (SeatN, Zone, Class, PID, IDNumber) VALUES ('1C', 2, 'Economy', 'P000002', 501); INSERT INTO BoardingPass (SeatN, Zone, Class, PID, IDNumber) VALUES ('3F', 1, 'Business', 'P000003', 502); INSERT INTO BoardingPass (SeatN, Zone, Class, PID, IDNumber) VALUES ('2H', 3, 'Economy', 'P000004', 503); INSERT INTO BoardingPass (SeatN, Zone, Class, PID, IDNumber) VALUES ('1D', 1, 'First', 'P000005', 505);</pre>                                                                                                                                                                                                                                                                                                                                                                                                                                     |      |          |         |          |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| Data       | <div><div>Script Output xQuery Result x</div><div> SQL   All Rows Fetched: 5 in 0.001 seconds</div><table><tr><th></th><th>SEATN</th><th>ZONE</th><th>CLASS</th><th>PID</th><th>IDNUMBER</th></tr><tr><td>1</td><td>2A</td><td>1</td><td>First</td><td>P000001</td><td>501</td></tr><tr><td>2</td><td>3F</td><td>1</td><td>Business</td><td>P000003</td><td>502</td></tr><tr><td>3</td><td>1D</td><td>1</td><td>First</td><td>P000005</td><td>505</td></tr><tr><td>4</td><td>1C</td><td>2</td><td>Economy</td><td>P000002</td><td>501</td></tr><tr><td>5</td><td>2H</td><td>3</td><td>Economy</td><td>P000004</td><td>503</td></tr></table></div> |      |          |         |          |  | SEATN | ZONE | CLASS | PID | IDNUMBER | 1 | 2A | 1 | First | P000001 | 501 | 2 | 3F | 1 | Business | P000003 | 502 | 3 | 1D | 1 | First | P000005 | 505 | 4 | 1C | 2 | Economy | P000002 | 501 | 5 | 2H | 3 | Economy | P000004 | 503 |
|            | SEATN                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | ZONE | CLASS    | PID     | IDNUMBER |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| 1          | 2A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1    | First    | P000001 | 501      |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| 2          | 3F                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1    | Business | P000003 | 502      |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| 3          | 1D                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1    | First    | P000005 | 505      |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| 4          | 1C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2    | Economy  | P000002 | 501      |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |
| 5          | 2H                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 3    | Economy  | P000004 | 503      |  |       |      |       |     |          |   |    |   |       |         |     |   |    |   |          |         |     |   |    |   |       |         |     |   |    |   |         |         |     |   |    |   |         |         |     |

## Queries

```
SELECT FromLocation, ToLocation
FROM Flights
WHERE FNumber=404;
```

## Result

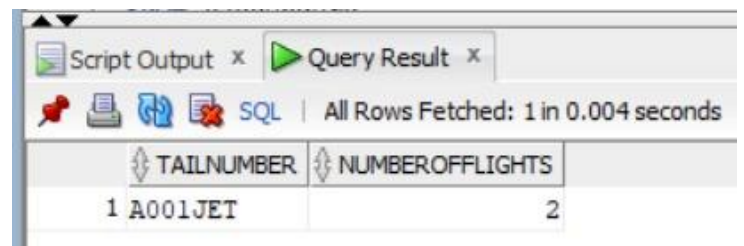


Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.002 seconds

|   | FROMLOCATION | TOLOCATION |
|---|--------------|------------|
| 1 | TX           | MI         |

```
SELECT TailNumber, COUNT(FNumber) AS NumberOfFlights
FROM Flights
GROUP BY TailNumber
HAVING
    COUNT(FNumber) >= 2;
```

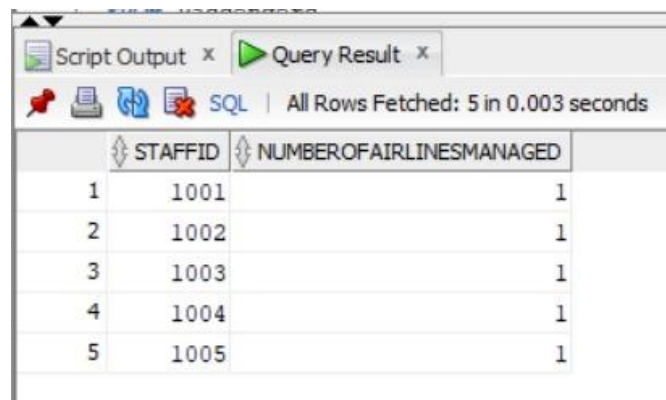


Script Output x Query Result x

SQL | All Rows Fetched: 1 in 0.004 seconds

|   | TAILNUMBER | NUMBEROFFLIGHTS |
|---|------------|-----------------|
| 1 | A001JET    | 2               |

```
SELECT StaffID,
    COUNT(IDNumber) AS NumberOfAirlinesManaged
FROM AirLines
GROUP BY StaffID
HAVING
    COUNT(IDNumber) = 1;
```



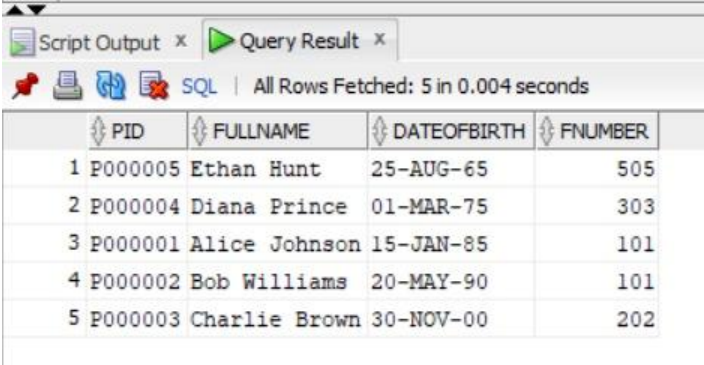
Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.003 seconds

|   | STAFFID | NUMBEROFAIRLINESMANAGED |
|---|---------|-------------------------|
| 1 | 1001    | 1                       |
| 2 | 1002    | 1                       |
| 3 | 1003    | 1                       |
| 4 | 1004    | 1                       |
| 5 | 1005    | 1                       |

```
SELECT PID,FullName, DateOfBirth, FNumber
FROM Passengers
ORDER BY DateOfBirth ASC;
```

```
SELECT BagTag,Weight, Status, PID
FROM
    Baggage
ORDER BY
    Weight DESC;
```

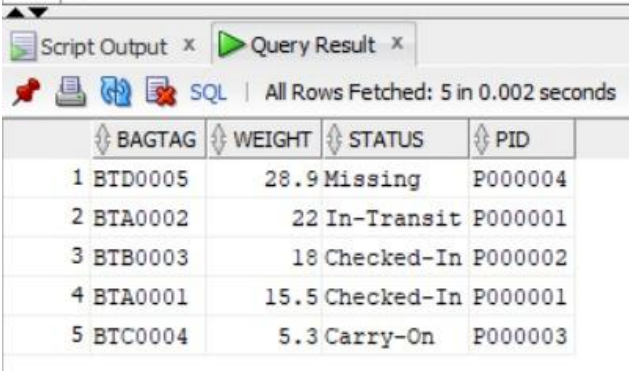


Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.004 seconds

|   | PID     | FULLNAME      | DATEOFBIRTH | FNUMBER |
|---|---------|---------------|-------------|---------|
| 1 | P000005 | Ethan Hunt    | 25-AUG-65   | 505     |
| 2 | P000004 | Diana Prince  | 01-MAR-75   | 303     |
| 3 | P000001 | Alice Johnson | 15-JAN-85   | 101     |
| 4 | P000002 | Bob Williams  | 20-MAY-90   | 101     |
| 5 | P000003 | Charlie Brown | 30-NOV-00   | 202     |

```
SELECT BagTag,Weight, Status, PID
FROM
    Baggage
ORDER BY
    Weight DESC;
```



Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.002 seconds

|   | BAGTAG  | WEIGHT | STATUS     | PID     |
|---|---------|--------|------------|---------|
| 1 | BTD0005 | 28.9   | Missing    | P000004 |
| 2 | BTA0002 | 22     | In-Transit | P000001 |
| 3 | BTB0003 | 18     | Checked-In | P000002 |
| 4 | BTA0001 | 15.5   | Checked-In | P000001 |
| 5 | BTC0004 | 5.3    | Carry-On   | P000003 |

```
SELECT P.FullName,T.Locationnn AS TerminalLocation,G.GateN
FROM Passengers P
INNER JOIN Gates G ON P.GateN = G.GateN
INNER JOIN Terminals T ON G.TerminalID = T.TerminalID;
```



Script Output x Query Result x

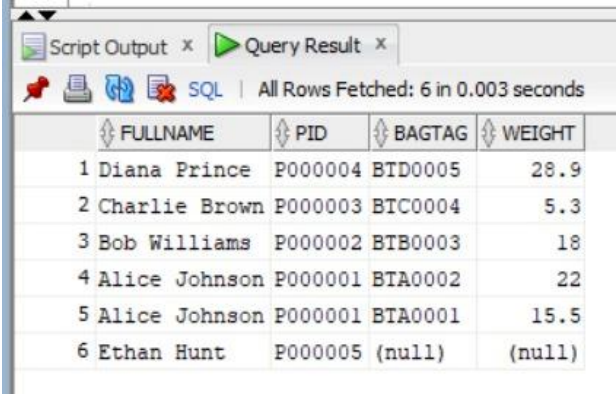
SQL | All Rows Fetched: 5 in 0.002 seconds

|   | FULLNAME      | TERMINALLOCATION | GATEN |
|---|---------------|------------------|-------|
| 1 | Charlie Brown | North Wing A     | 2     |
| 2 | Diana Prince  | South Wing B     | 10    |
| 3 | Alice Johnson | International C  | 11    |
| 4 | Bob Williams  | International C  | 11    |
| 5 | Ethan Hunt    | Domestic D       | 20    |

```

SELECT P.FullName, P.PID, B.BagTag, B.Weight
FROM Passengers P
LEFT JOIN Baggage B ON P.PID = B.PID
ORDER BY B.BagTag DESC NULLS LAST;

```



Script Output x Query Result x

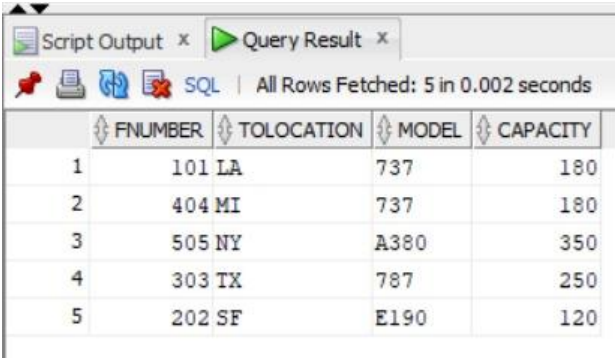
SQL | All Rows Fetched: 6 in 0.003 seconds

|   | FULLNAME      | PID     | BAGTAG   | WEIGHT |
|---|---------------|---------|----------|--------|
| 1 | Diana Prince  | P000004 | BTID0005 | 28.9   |
| 2 | Charlie Brown | P000003 | BTC0004  | 5.3    |
| 3 | Bob Williams  | P000002 | BTB0003  | 18     |
| 4 | Alice Johnson | P000001 | BTA0002  | 22     |
| 5 | Alice Johnson | P000001 | BTA0001  | 15.5   |
| 6 | Ethan Hunt    | P000005 | (null)   | (null) |

```

SELECT FNumber, ToLocation, Model, Capacity
FROM Flights
NATURAL JOIN
AirPlane;

```



Script Output x Query Result x

SQL | All Rows Fetched: 5 in 0.002 seconds

|   | FNUMBER | TOLOCATION | MODEL | CAPACITY |
|---|---------|------------|-------|----------|
| 1 | 101     | LA         | 737   | 180      |
| 2 | 404     | MI         | 737   | 180      |
| 3 | 505     | NY         | A380  | 350      |
| 4 | 303     | TX         | 787   | 250      |
| 5 | 202     | SF         | E190  | 120      |

# Chapter 5: Conclusion

## 5.1 Conclusion

This project aimed to build a airport management database system to help facilitate accurate and efficient data storage and retrieval and improve overall operational efficiency within airport environment.

We started by analyzing the system requirements and identifying the main entities and their attributes and relationships among them, created the conceptual model (EER Diagram) and converted it into relational model to show how the data is connected in tables with primary and foreign keys.

Finally, we implemented the database using SQL to create the tables with appropriate data types and integrity constraints, insert sample data, and execute several queries to test data retrieval, aggregation, sorting and joining between multiple tables .

## 5.2 Lessons Learn

- Learned to work in a team and divide tasks efficiently.
- Learned how to analyze a real-world system and identify its main entities, attributes, and relationships.
- Learned how to design a conceptual model and map it to relational model .
- Learned to build and implement a database and execute queries using SQL.

## 5.1 Conclusion

This project aimed to build a airport management database system to help facilitate accurate and efficient data storage and retrieval and improve overall operational efficiency within airport environment.

# References

- [1] R. Elmasri, FUNDAMENTALS OF Data Types, Pearson, 2007.